The IQ 150 thought piece

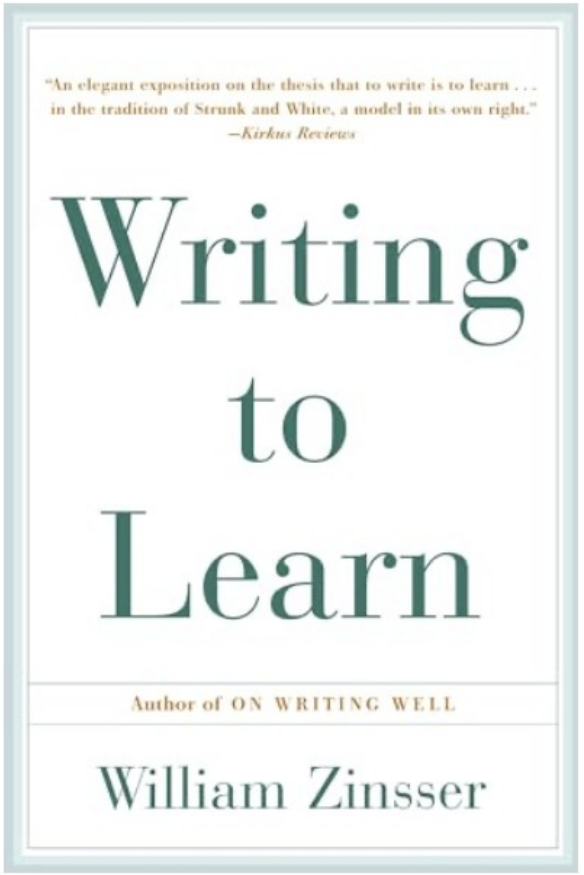
What scientific task would this label help us predict?

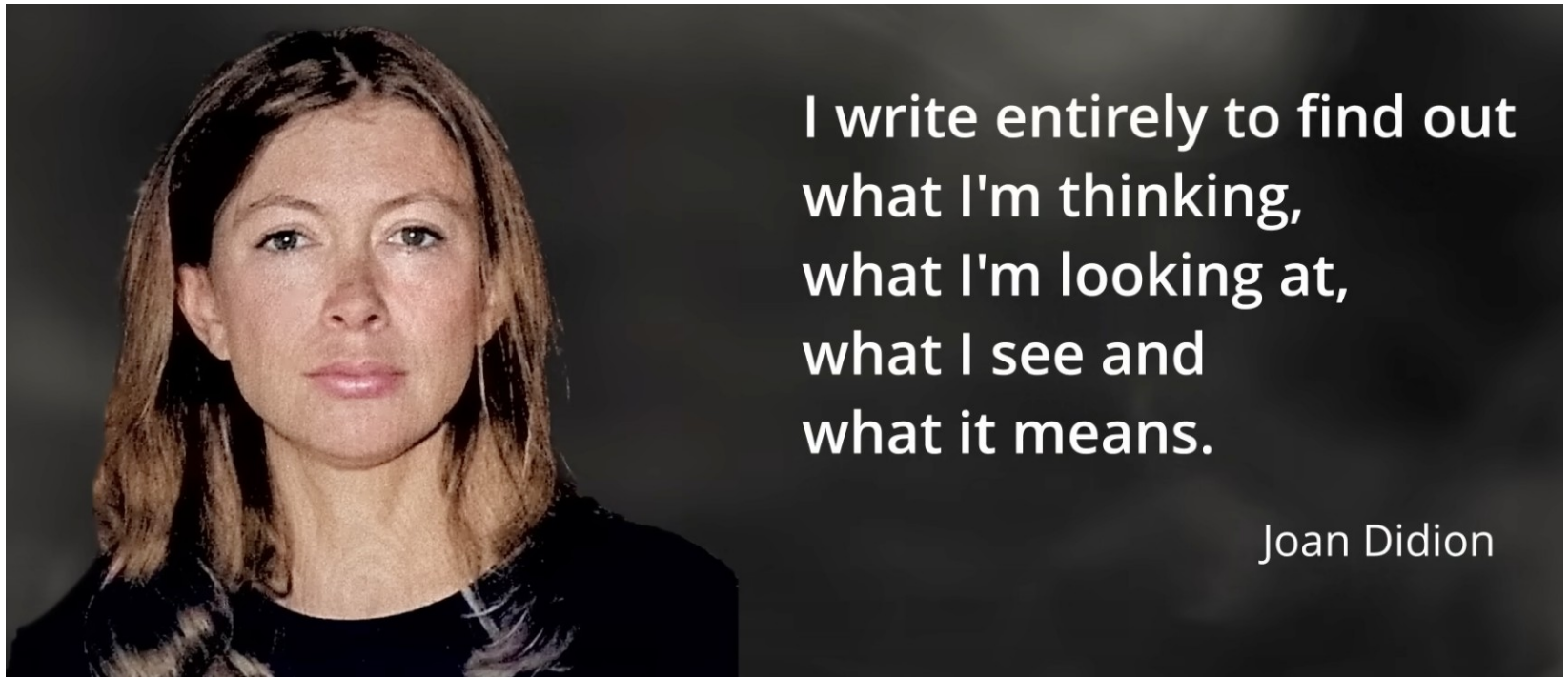
How do test exposure, tools and time budgets affect a score?

Which failures remain invisible to a single aggregate number?

Propose a capability profile and an experiment that could refute it.

I strongly encourage you to journal this class





AI review of the course materials

Audit every lecture, note, paper link, assignment and rubric.

Check factual claims, mathematical notation and executable examples.

Record source, version, issue, proposed correction and reviewer.

An instructor resolves uncertain claims and approves material for release.

A repository for students and agents

Each module includes lecture notes and curated background papers.

Assignments ship code, tests, run instructions and evaluation rubrics.

Thought pieces connect assumptions to evidence and counterexamples.

Challenge records state the question, baseline, budget and success test.

Six labs and a project

Lab 0 Coding-loop setup and backend smoke tests

Labs 1–2 Einstein notation, attention variants and multimodal scaling

Labs 3–4 Serving, concurrency, benchmarking and Jev decisions

Lab 5 Scientific replication and a bounded Navier–Stokes task

Assessment with agents available

50% labs Correctness, experiments and reproducible evidence

40% project Question, implementation, validation and oral defense

10% participation Thought pieces, reviews and the learning journal

Students explain and defend their own submitted work.

The ATLAS of problems

Start with the reviewed AI ATLAS edition dated 18 September 2026.

100 entries cover scientific questions, methods and verification projects.

Choose a tractable subproblem with data and an independent success test.

Keep the original scientific ambition and state the limits of the result.

Navier–Stokes as a bounded challenge

Define the PDE, domain, boundary conditions and initial data.

Compare a numerical baseline with a learned surrogate.

Evaluate held-out trajectories, rollout stability and physical constraints.

Separate numerical error, modeling error and distribution shift.

Models that are facile with theory and can create and debate alternative mechanisms

3. **Theory formation and mechanistic explanation**

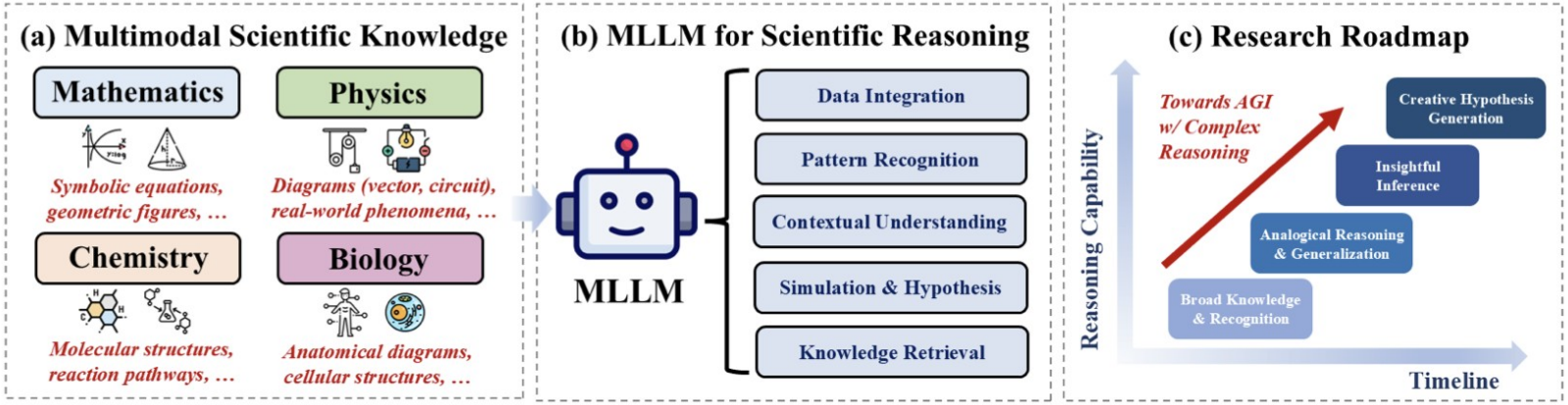
- **What:** Integrations of neural reasoning, program synthesis, and formal methods that propose *explanatory* models (laws, symmetries, invariants), not just predictive fits.
- **Why:** Scientific value is explanation that compresses data and predicts out-of-distribution.
- **Good looks like:** Auto-generated, **machine-checkable** theories that (i) pass formal proof checks where applicable and (ii) make novel, confirmed predictions. Progress in autoformalization/theorem-proving is an existence proof of the tooling direction. arXiv +2

Combining current state of the art reasoning models with theorem provers (Lean4, Coq, etc) and symbolic computing engines (sympy sandbox) gets us part way to this goal. Specialized models that are trained to leverage these tools might do even better in the near term.

Multimodal scientific reading and reasoning models that understand provenance and time

4. **Multimodal scientific reading with provenance**

- **What:** Robust ingestion of papers, figures, tables, methods, code, and data; automatic replication; citation-level provenance and uncertainty.
- **Why:** Mastery of literature must be *verifiable* and *faithful*, not a paraphrase generator.
- **Good looks like:** Reproduce ≥80% of computational results for a representative set of papers (code+data), generate discrepancy reports with exact citations, and attach confidence to every claim.



Thought pieces throughout the course

When does a verifier deserve trust?

What does a faster or larger system actually explain?

Can agent memory and concurrency produce reliable cumulative progress?

What evidence would distinguish discovery from plausible synthesis?

The project evidence package

A precise question and a budgeted experimental plan

A versioned implementation with baselines and ablations

Data provenance, raw results, failure analysis and reproducible commands

A short paper, poster and oral defense

Outcomes we are driving towards

- **An AI that proposes a theory**
  not just a fit to data, but a falsifiable mechanism that predicts new phenomena and survives experimental test.

- **An AI that designs and runs its own experiments**
  closing the loop from hypothesis to instrument control to result interpretation to next experiment.

- **An AI that reliably says 'I don't know'**
  with calibrated uncertainty that scientists trust enough to act on — including knowing when not to act.

- **Scientific AI as public infrastructure**
  open models, shared compute, common standards — so every lab in every country can participate.

- **Incentive reform**
  where replication, negative results, data curation, and maintenance are valued as highly as novel discoveries.

The first week

Read the course guide and create a learning journal.

Complete the environment and backend checks.

Run a deliberately failing example, repair it with an agent and verify it.

Submit the patch, test evidence and a short account of what failed.

Questions that will guide the quarter

Which capabilities can we verify independently?

How does systems design change the capability we observe?

Where does additional scale stop helping?

What would count as a scientific contribution?

REFERENCE

Selected sources

2025 foundation LS-2025-First-Lecture

Scientific systems Purdue September 2026 and Scientific Superintelligence

Coding loops Closing the Loop pi-30 tutorial, July 2026

Planning and challenges CODE-A-THON-RICK and reviewed AI ATLAS